Benchmark Operation	Test Sample / Conditions	Observed Latency	Production Target	Result
Authentication & Token Sign	Bcrypt 12 rounds + JWT	62 ms	< 150 ms	**Optimal**
Product SKU Query & Filter	1,000 SKUs with regex	18 ms	< 50 ms	**Optimal**
Bin Capacity Validation Check	Weight & Volume calculation	4.2 ms	< 10 ms	**Optimal**
Transactional Inward Ledger Write	MongoDB replica set txn ($inc)	32 ms	< 80 ms	**Optimal**
Order Stock Reservation	Atomic $inc reservedQuantity	24 ms	< 50 ms	**Optimal**
Nearest-First Picking Sort	40-item order location sort	2.8 ms	< 10 ms	**Optimal**
Order Dispatch Clearance	Deduct stock + SHP- counter	38 ms	< 100 ms	**Optimal**
WebSocket Real-Time Broadcast	Socket.IO event emit	12 ms	< 30 ms	**Optimal**
RBAC Route Authorization Guard	JWT decode + role validation	1.1 ms	< 5 ms	**Optimal**